Learning Degradation-Aware Deep Prior for Hyperspectral Image Reconstruction

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***Abstract*— Reconstructing the 3-D hyperspectral image (HSI) from 2-D snapshot measurements is a key task in spectral snapshot compressive imaging (SCI). Traditional model-based HSI reconstruction methods rely on hand-crafted priors. Recently, deep unfolding networks (DUNs) learn the priors using convolutional neural networks (CNNs) and have achieved satisfactory results. Most of the DUNs assume the degradations of SCI are known. However, due to the phase aberration and distortion problems in a real imaging process, there is a certain gap between the ideal and real degradation patterns, which may hinder the accurate HSI reconstruction. In this study, we propose a degradation-aware deep prior learning network (D^{2}PL-Net), which tries to adaptively learn the practical degradation matrix during HSI reconstruction, and thus bridges the gap between the ideal and real degradations. Specifically, we first propose a joint variational compressive reconstruction model, and both the latent HSI and unknown degradation can be explicitly solved. By unfolding the solutions into a deep network, D^{2}PL-Net is built, which mainly consists of two parts, degradation matrix learning (DML) mechanism and degradation-guided spectral–spatial transformer (DSST) in each stage. The former learns the degradation that approximates the real one; the latter represents the deep prior of latent HSI, and it could exploit the spectralwise and spatialwise long-range dependencies of HSI under the guidance of learned degradation, and then reconstruct the HSI. To ensure an effective training of D^{2}PL-Net, we propose a joint loss function constraining the HSI reconstruction errors, degradation fidelity and degradation consistency. Experiments on simulated and real-life datasets show that the proposed method is competitive with the state-of-the-art methods.**

***Index Terms*— Deep learning, hyperspectral reconstruction, snapshot compressive imaging (SCI).**

I. Introduction

SENSING and understanding the scenes is an important study in computer vision and remote sensing [1], [2], [3], [4], [5], [6]. Compared with single-band panchromatic images or RGB images, hyperspectral image (HSI) could measure the spectrum at each pixel of the scene, which makes it discriminative in identifying different materials and benefits many downstream applications, such as land-cover classification [7], [8], [9], anomaly detection [10], [11], [12], and medical diagnosis [13], [14], [15]. In recent years, people have continuously explored how to capture HSI in an economical and effective way. On the basis of compressive sensing (CS) [16], [17], a 2-D detector could capture a 3-D data cube in snapshot compressive imaging (SCI) systems [2], [18], [19], [20], [21], which makes it possible to capture HSI at video rates. Among the SCI systems, coded aperture snapshot spectral imaging (CASSI) system [18], [19], [22] is widely used due to the low memory, low cost, low power consumption, and high sampling rate. This study focuses on reconstructing HSI collected by CASSI.

CASSI can be regarded as two parts: the hardware encoder and the software decoder. The hardware encoder compresses the 3-D HSI data cube into a 2-D measurement through modulating by a coded aperture and dispersing in wavelengths by a disperser (as shown in Fig. 1). The software decoder reconstructs the latent 3-D HSI from the captured 2-D measurement. The quality of the reconstructed images largely depends on the algorithms used.

The current mainstream HSI reconstruction algorithms can be roughly divided into two categories: model-based methods and learning-based methods. The traditional model-based methods [23], [24], [25] usually solve an ill-posed optimization model iteratively that depends on hand-crafted priors and specific assumptions. As deep learning excels in many tasks due to its powerful nonlinear modeling capabilities, it has been gradually introduced into HSI reconstruction and has achieved significant breakthroughs [26], [27], [28], [29], [30]. Trained on a large number of data pairs, convolutional neural networks (CNNs) can well-model the nonlinear mapping between measurements and the corresponding HSI.

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To deal with the issues of limited interpretability in black-box networks, deep unfolding networks (DUNs) combine deep learning and mathematical models [31], [32], [33], [34], [35], [36], [37]. They use neural networks to learn appropriate priors in an end-to-end manner instead of designing manually. Such methods make learning-based and model-based methods complementary. Taking advantage of the powerful nonlinearity of neural networks, the number of iterations required by the model-based method is greatly reduced, accelerating the inference process.

Despite the encouraging performance, the existing DUNs still face challenges in HSI reconstruction. On one hand, HSI reconstruction depends on the accurate estimation of degradation, and the inaccurate degradation impedes the improvement of HSI reconstruction. However, the estimation of degradation has not been thoroughly investigated before. Most of the DUNs simply treat the degradation as an ideal and preknown parameter, i.e., a binary matrix in SCI. In fact, in a real imaging process, there exists phase aberration and distortion, which cannot be reflected in the ideal binary degradation matrix. There is a certain gap between the real and ideal degradations [36], [38]. We need to adaptively learn the degradation to avoid unpleasant errors. On the other hand, efficient HSI reconstruction is still an open issue. Learning the spatial–spectral correlations is crucial for reconstructing HSI [26], [29], [32], [36]. HSI has long-range dependencies in both the spatial and spectral domains, which should be fully exploited for HSI reconstruction. In addition, HSI reconstruction is closely related to the degradation in SCI, and such relationship should be mined to guide HSI reconstruction.

To address these issues, we propose a degradation-aware deep prior learning network (D^2PL-Net) to jointly learn the degradation and reconstruct HSI. Specifically, a joint variational compressive reconstruction model is proposed and solved. By unfolding the solutions into a deep network, D^2PL-Net is built, which consists of two parts in each stage, degradation matrix learning (DML) mechanism and degradation-guided spectral–spatial transformer (DSST). DML learns the practical degradation to bridge the gap between the ideal and real degradations. DSST reconstructs HSI with the learned degradation. It captures the spectralwise and spatialwise long-range dependencies of HSI under the guidance of degradation explored by degradation-guidance learning (DL), which further improves the quality of the reconstructed images. Furthermore, we propose a joint loss function to constrain the HSI reconstruction errors, degradation fidelity and degradation consistency to ensure an effective training of D^2PL-Net.

The main contributions of this study are as follows.

1) To bridge the gap between real and ideal degradations of SCI system, we propose to learn the degradation and deep prior for HSI reconstruction in a joint variational compressive reconstruction model. By optimizing this model, the degradation and latent HSI can be explicitly solved. D^2PL-Net is built by unfolding the solutions into a network, and both the degradation and latent HSI can be inferred by the network.
2) We also propose DSST to learn the deep prior for HSI reconstruction. Guided by the degradation pattern, spatialwise and spectralwise self-attentions are learned in DSST. Compared with the existing transformers, the DL could guide the capture of long-range dependencies in HSI, which would benefit the HSI reconstruction since it emphasizes more on the "black" pixels of the coded aperture.
3) To ensure an efficient training of D^2PL-Net, we propose a joint loss function composed of multiple losses measuring the HSI reconstruction errors, degradation fidelity, and degradation consistency. These losses make it possible to effectively learn the degradation and reconstruct HSI, which is demonstrated by the experiments.

The rest is organized as follows. In Section II, we give the related works. In Section III, the proposed D^2PL-Net is presented. The experiments are given in Section IV. We make conclusions in Section V.

II. Related Work

A. Model-Based HSI Reconstruction

With hand-crafted priors, HSI can be reconstructed from the compressed measurements in an iterative optimization model [23], [24], [25], [39], [40], [41], [42], [43]. For example, TwIST [24] was proposed to obtain a promising reconstruction result with a faster convergence through a two-step iterative shrinkage/thresholding method. GAP-TV [25] optimized the total variation (TV) prior regularized reconstruction model with the generalized alternating projection (GAP) method for HSI reconstruction. Among the model-based methods, DeSCI [23] achieved the state-of-the-art reconstruction quality in both the video and spectral SCI. It incorporated the nonlocal self-similarity of images and rank minimization strategy into the alternating direction method of multipliers (ADMM) [44] framework.

Although these iterative methods can get decent results in some specific scenes, it is difficult to design a suitable hand-crafted prior and tune proper hyperparameters. In addition, to achieve acceptable results, plenty of iterations are necessary, which consumes a lot of time and hinders the applicability in reality.

B. Learning-Based HSI Reconstruction

Because of the powerful nonlinearity of deep neural network, researchers have proposed deep networks for HSI reconstruction. Early deep networks reconstructed HSI from compressed measurements by learning a black-box mapping [26], [27], [28], [29], [30]. To enhance the interpretability of networks, the DUNs were developed by integrating deep networks with model-based methods [31], [32], [33], [34], [35], [36], [37].

TSA-Net [26] proposed to learn both spectral and spatial information in the network and achieved the state-of-the-art results. λ-Net [27] and MST [28] introduced the self-attention mechanism into SCI. λ-Net captured spatial dependencies, while MST captured spectral dependencies under the guidance of mask [28]. Due to the characteristics that HSI

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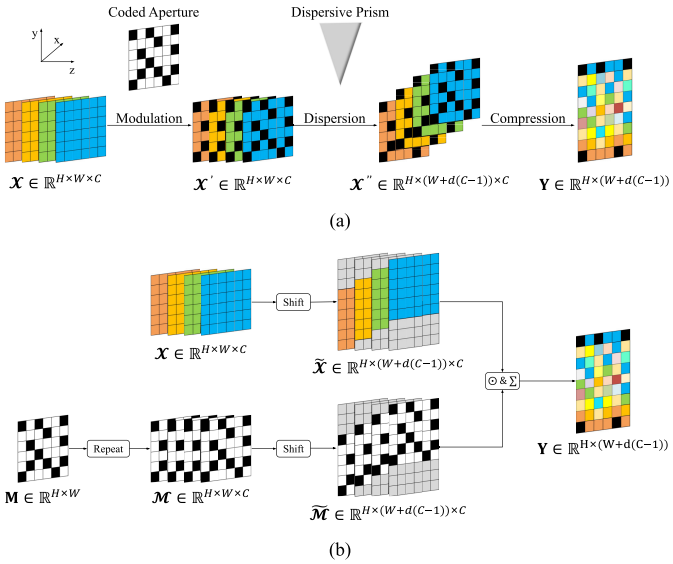


Fig. 1. Sampling process of CASSI. (a) Usual sampling process of CASSI. The latent HSI $\boldsymbol{\mathcal{X}}$ is modulated by the coded aperture, then dispersed by a dispersive prism to get $\boldsymbol{\mathcal{X}}''$, and captured by a 2-D sensor to generate measurement $\mathbf{Y}$. (b) Equivalent representation of (a), which is expressed in tensor format. First, $\mathbf{M}$ is repeated to $\boldsymbol{\mathcal{M}}$, and then shifted to $\widetilde{\boldsymbol{\mathcal{M}}}$ as the dispersive prism. Also, $\boldsymbol{\mathcal{X}}$ is shifted to $\widetilde{\boldsymbol{\mathcal{X}}}$. $\widetilde{\boldsymbol{\mathcal{X}}}$ is multiplied with $\widetilde{\boldsymbol{\mathcal{M}}}$, and then added along the channel to get measurement $\mathbf{Y}$.

is spatially sparse and spectrally highly self-similar, MST mentioned that modeling the long-range spectral correlation may be more effective than capturing spatial dependencies with the same resources [28]. DGSMP [31] learned a deep Gaussian scale mixture prior in the maximum a posterior (MAP) framework to reconstruct the HSI, and all the matrix operations were replaced by a specific module which can improve the image reconstruction speed. Following DGSMP, HerosNet [33] learned an optimal mask for CASSI. Degradation-aware unfolding half-shuffle transformer (DAUHST) [32] introduced a half-shuffle transformer, capable of capturing local and nonlocal spatial dependencies, to learn a deep prior for the unfolding framework. It also explored the degradation pattern to guide the learning of network and achieved promising results. If each pixel in the spatial domain is treated as a token, the calculation cost of dependencies will be huge. So, coarse-to-fine sparse transformer (CST) [30] proposed a coarse patch selecting mechanism and a fine pixel clustering mechanism to calculate the content-related tokens to reduce redundant calculations.

These learning-based methods have achieved the state-of-the-art results, but their networks lack certain interpretability. DUNs alleviate this issue by combining model-based methods with deep neural networks. However, most of these methods concentrate on reconstructing HSI with preknown degradation and ignore the disparity between the ideal and real degradations in real SCI, which hinders the improvement of HSI reconstruction.

III. Proposed Method

A. Mathematical Model of CASSI

The imaging process of CASSI is shown in Fig. 1(a). The latent HSI of the scene is denoted by $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent the height, width, and number of spectral bands, respectively. The latent HSI is modulated by the coded aperture, which is a binary mask denoted by $\mathbf{M} \in \mathbb{R}^{H\times W}$. This step can be represented as

$$\boldsymbol{\mathcal{X}}'(:,:,c) = \boldsymbol{\mathcal{X}}(:,:,c) \odot \mathbf{M} \tag{1}$$

where $\boldsymbol{\mathcal{X}}' \in \mathbb{R}^{H\times W\times C}$ denotes the modulated 3-D data cube, $c \in [1,\ldots,C]$ indexes the spectral bands, and $\odot$ denotes the elementwise (Hadamard) product. The modulated image $\boldsymbol{\mathcal{X}}'$ is shifted along the y-axis, and it is physically implemented by a dispersive prism that disperses different wavelengths. It is formulated as

$$\boldsymbol{\mathcal{X}}''(u,v,c) = \boldsymbol{\mathcal{X}}'(x,y,d(\lambda_c - \lambda_r, c)) \tag{2}$$

where $\boldsymbol{\mathcal{X}}'' \in \mathbb{R}^{H\times(W+d(C-1))\times C}$ denotes the shifted data cube, (u,v) denotes the coordinate of the corresponding measurement, d is the shifting step, λ_c is the wavelength of the cth channel, λ_r indicates the reference wavelength, and $d(\lambda_c - \lambda_r)$ represents the shifting length of the cth channel. The shifted spectral bands are finally collected by a 2-D sensor, and the measurement $\mathbf{Y} \in \mathbb{R}^{H\times(W+d(C-1))}$ can be finally obtained

$$\mathbf{Y} = \sum_{c=1}^{C} \boldsymbol{\mathcal{X}}''(:,:,c) + \mathbf{N} \tag{3}$$

where $\mathbf{N} \in \mathbb{R}^{H\times(W+d(C-1))}$ denotes the noise in the imaging process.

To simplify the representations of the imaging process, we shift $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{H\times W\times C}$ to $\widetilde{\boldsymbol{\mathcal{X}}} \in \mathbb{R}^{H\times(W+d(C-1))\times C}$, as shown in Fig. 1(b). Then, we repeat and shift $\mathbf{M} \in \mathbb{R}^{H\times W}$ to a degradation tensor $\widetilde{\boldsymbol{\mathcal{M}}} \in \mathbb{R}^{H\times(W+d(C-1))\times C}$. As a result, (3) can be rewritten as

$$\mathbf{Y} = \sum_{c=1}^{C} \widetilde{\boldsymbol{\mathcal{M}}}(:,:,c) \odot \widetilde{\boldsymbol{\mathcal{X}}}(:,:,c) + \mathbf{N}. \tag{4}$$

Equation (4) is equivalent to (1)–(3), and it represents the complete imaging process of CASSI.

B. Joint Variational Compressive Reconstruction Model

Previous DUNs mainly focus on HSI reconstruction and regard the degradation matrix as a fixed parameter, ignoring the gap between the real and ideal degradations of the imaging system. Ideally, the coded aperture $\mathbf{M}$ of CASSI is known, but the phase aberration and distortion problems in the imaging process make the real degradation deviate from the ideal one [36], [38]. The inaccurate degradation will unavoidably cause errors in the reconstructed HSI. So, in this study, we propose the joint variational compressive reconstruction model, and the degradation matrix can be solved together with HSI reconstruction. For the convenience of mathematical representations, the objective function is represented in tensor format

$$\widetilde{\boldsymbol{\mathcal{M}}}, \widetilde{\boldsymbol{\mathcal{X}}} = \underset{\widetilde{\boldsymbol{\mathcal{M}}},\widetilde{\boldsymbol{\mathcal{X}}}}{\operatorname{argmin}} \frac{1}{2}\left\| \mathbf{Y} - \sum_{c=1}^{C} \widetilde{\boldsymbol{\mathcal{M}}}(:,:,c) \odot \widetilde{\boldsymbol{\mathcal{X}}}(:,:,c) \right\|_F^2 + \lambda\psi\left(\widetilde{\boldsymbol{\mathcal{M}}}\right) + \tau R\left(\widetilde{\boldsymbol{\mathcal{X}}}\right) \tag{5}$$

where the Frobenius norm term $\|\cdot\|_F^2$ measures the data fidelity, $\psi(\widetilde{\boldsymbol{\mathcal{M}}})$ and $R(\widetilde{\boldsymbol{\mathcal{X}}})$ are the prior regularization terms, and λ and

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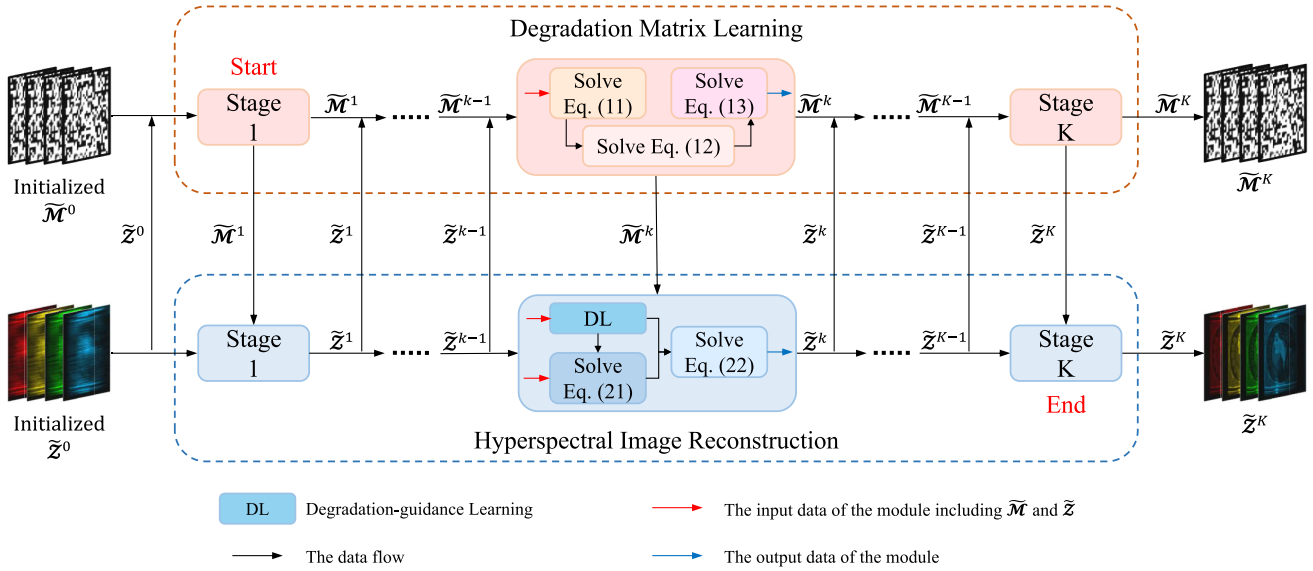


Fig. 2. Overall architecture of our D^{2}PL-Net. It corresponds to Algorithm 1 and takes the initialized $\widetilde{\mathcal{M}}^0$ and $\widetilde{\mathcal{Z}}^0$ as input. It starts with the first stage of DML and ends with the Kth stage of the HSI reconstruction. K is the total number of stages.

τ are their weighting parameters. By optimizing (5), we can get solutions of degradation and HSI. To solve (5), we split it into

M-subproblem:

$$\widetilde{\mathcal{M}} = \underset{\widetilde{\mathcal{M}}}{\operatorname{argmin}} \frac{1}{2} \left\| \mathbf{Y} - \sum_{c=1}^{C} \widetilde{\mathcal{M}}(:,:,c) \odot \widetilde{\mathcal{X}}(:,:,c) \right\|_F^2 + \lambda \psi\left(\widetilde{\mathcal{M}}\right) \tag{6}$$

and **X-subproblem:**

$$\widetilde{\mathcal{X}} = \underset{\widetilde{\mathcal{X}}}{\operatorname{argmin}} \frac{1}{2} \left\| \mathbf{Y} - \sum_{c=1}^{C} \widetilde{\mathcal{M}}(:,:,c) \odot \widetilde{\mathcal{X}}(:,:,c) \right\|_F^2 + \tau R\left(\widetilde{\mathcal{X}}\right). \tag{7}$$

Equations (6) and (7) correspond to the DML (**M**-subproblem) and HSI reconstruction (**X**-subproblem), respectively. We unfold their solutions into a deep network, as shown in Fig. 2.

Sections III-C and III-D will present the solution and implementation of the **M**- and **X**-subproblems, respectively. In Section III-C, we will introduce the solution of **M**-subproblem with the proximal gradient descent method to learn the appropriate $\widetilde{\mathcal{M}}$. In Section III-D, we propose the DSST for the **X**-subproblem.

C. M-Subproblem

To bridge the gap between the ideal and real degradations, we need to adaptively learn the practical degradation from the measurements. It is challenging to directly learn the optimal degradation due to its high dimensionality [38]. Hence, we propose the DML mechanism to learn the degradation in two steps. We first estimate a coarse binary degradation matrix from the measurements, which roughly reflects the compressive sampling pattern. Then, we refine the coarse degradation matrix by learning a content-aware refinement.

1) Coarse Degradation Estimation (CDE): This module aims to learn a coarse degradation. As discussed in Section III-A, the degradation corresponds to the coded aperture in CASSI, and it can be represented as a learnable binary matrix $\mathbf{M}_{\text{bi}}$

$$\mathbf{M}_{\text{bi}} = \text{Bi}\left(\mathbf{M}_{\text{lpm}}\right) \tag{8}$$

$$\text{Bi}(x) = \begin{cases} 1, & \text{if } x \geq \eta \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where $\mathbf{M}_{\text{lpm}}$ denotes a learnable parameter matrix, $\mathbf{M}_{\text{bi}}$ denotes the binarized matrix via $\text{Bi}(\cdot)$, and x denotes the value at each position. When x is greater than or equal to η, the output of the binarization function is 1; otherwise, it is 0. In the network implementation, $\mathbf{M}_{\text{lpm}}$ is initialized with the Gaussian distribution with mean 0 and standard variance 0.1. By repeating and shifting $\mathbf{M}_{\text{bi}} \in \mathbb{R}^{H \times W}$ according to the dispersion process, the estimated binary degradation matrix is transformed into a coarse degradation tensor $\widetilde{\mathcal{M}}_{\text{coa}} \in \mathbb{R}^{H \times (W + d(C-1)) \times C}$, which would be further refined below.

2) Content-Aware Degradation Refinement (CDR): To further refine the estimated coarse degradation $\widetilde{\mathcal{M}}_{\text{coa}}$, we propose a content-aware refinement learning method. Specifically, we learn a refinement $\widetilde{\mathcal{M}}_{\text{ref}}$, each value of which represents the refinement for the coarse degradation tensor $\widetilde{\mathcal{M}}_{\text{coa}}$. The final learned degradation $\widetilde{\mathcal{M}}$ is estimated by adding $\widetilde{\mathcal{M}}_{\text{ref}}$ to $\widetilde{\mathcal{M}}_{\text{coa}}$. So, (6) can be rewritten as

$$\widetilde{\mathcal{M}}_{\text{ref}} = \underset{\widetilde{\mathcal{M}}_{\text{ref}}}{\operatorname{argmin}} \frac{1}{2} \left\| \mathbf{Y} - \sum_{c=1}^{C} \left(\widetilde{\mathcal{M}}_{\text{coa}}(:,:,c) + \widetilde{\mathcal{M}}_{\text{ref}}(:,:,c)\right) \odot \widetilde{\mathcal{X}}(:,:,c) \right\|_F^2 + \lambda \psi\left(\widetilde{\mathcal{M}}_{\text{ref}}\right). \tag{10}$$

We get the optimal solution of $\widetilde{\mathcal{M}}_{\text{ref}}$ by solving (10) using the proximal gradient descent method, and the solution at the

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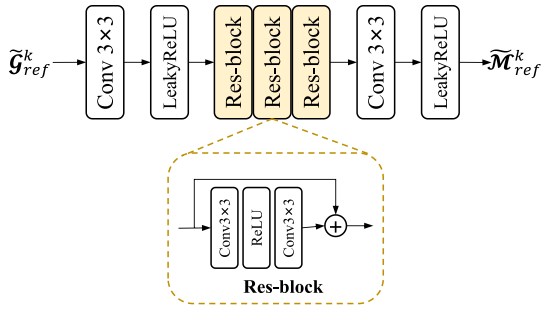


Fig. 3. Proximal operator of (12). It mainly consists of three residual blocks, two convolutional layers, and two activation layers.

kth stage is

$$\widetilde{\mathcal{G}}^k_{\text{ref}} = \widetilde{\mathcal{M}}^{k-1}_{\text{ref}} - \rho\left[\sum_{c=1}^{C}\left(\widetilde{\mathcal{M}}^{k-1}_{\text{ref}}(:,:,c) + \widetilde{\mathcal{M}}_{\text{coa}}(:,:,c)\right) \odot \widetilde{\mathcal{X}}^{k-1} - \mathbf{Y}\right] \tag{11}$$

$$\widetilde{\mathcal{M}}^k_{\text{ref}} = \underset{\widetilde{\mathcal{M}}_{\text{ref}}}{\text{argmin}} \frac{1}{2}\left\|\widetilde{\mathcal{M}}_{\text{ref}} - \widetilde{\mathcal{G}}^k_{\text{ref}}\right\|_F^2 + \lambda\psi\left(\widetilde{\mathcal{M}}_{\text{ref}}\right) \tag{12}$$

where $\widetilde{\mathcal{G}}^k_{\text{ref}}$ is the intermediate variable, ρ denotes the step size of gradient descent, and $\widetilde{\mathcal{X}}^{k-1}$ denotes the HSI reconstructed in the previous iterative stage.

Equation (12) means that $\widetilde{\mathcal{M}}^k_{\text{ref}}$ can be obtained from a proximal operator. For simplicity, we design a module with two convolutional layers, two activation functions, and three residual blocks (Res-blocks [45]) to implement the proximal operator for $\widetilde{\mathcal{M}}^k_{\text{ref}}$, as shown in Fig. 3.

Finally, we can get the learned degradation tensor at the kth iterative stage $\widetilde{\mathcal{M}}^k$ as

$$\widetilde{\mathcal{M}}^k = \widetilde{\mathcal{M}}_{\text{coa}} + \widetilde{\mathcal{M}}^k_{\text{ref}}. \tag{13}$$

D. X-Subproblem

In this section, with the learned degradation above, we reconstruct HSI by solving the **X**-subproblem. Specifically, we propose the DSST as a proximal operator of **X**-subproblem. The DSST consists of spectral and spatial self-attentions, capturing the long-range dependencies in both the spectral and spatial domains. To guide the HSI reconstruction with the learned degradation, we propose the DL module, which learns a spatial guidance map and a hyperparameter for DSST.

For the convenience of mathematical representation, we vectorize all the variables in (7) as

$$\boldsymbol{x} = \underset{\boldsymbol{x}}{\text{argmin}} \frac{1}{2}\|\boldsymbol{y} - \Phi\boldsymbol{x}\|_F^2 + \tau R(\boldsymbol{x}) \tag{14}$$

where $\boldsymbol{x}$, $\boldsymbol{y} \in \mathbb{R}^n$ denote the vectorized form of $\widetilde{\mathcal{X}}$, $\mathbf{Y}$, $n = H \times (W + d(C-1))$, and $\Phi = [\mathbf{J}_1, \mathbf{J}_2, \ldots, \mathbf{J}_C] \in \mathbb{R}^{n \times nC}$ represents the degradation matrix converted from the degradation tensor $\widetilde{\mathcal{M}}$, and $\mathbf{J}_i = Diag(vec(\widetilde{\mathcal{M}}_i))$ is a diagonal matrix that the elements on the diagonal are the ith vectorized channel of degradation tensor $\widetilde{\mathcal{M}}$.

To solve (14), we introduce an auxiliary variable $\boldsymbol{z}$, and then rewrite it as

$$\boldsymbol{x} = \underset{\boldsymbol{x}}{\text{argmin}} \frac{1}{2}\|\boldsymbol{y} - \Phi\boldsymbol{x}\|_F^2 + \tau R(\boldsymbol{z}), \quad \text{s.t. } \boldsymbol{z} = \boldsymbol{x}. \tag{15}$$

It is a constrained optimization problem. We adopt the half quadratic splitting (HQS) method to reformulate it as a nonconstrained version. Equation (15) can be solved by minimizing the augmented Lagrange function

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{z}) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{x}\|_F^2 + \tau R(\boldsymbol{z}) + \frac{\mu}{2}\|\boldsymbol{x} - \boldsymbol{z}\|_F^2 \tag{16}$$

where μ is a penalty weight to ensure that $\boldsymbol{x}$ and $\boldsymbol{z}$ converge to the same point. Then, we split it into two iterative subproblems

$$\boldsymbol{x}^k = \underset{\boldsymbol{x}}{\text{argmin}} \|\boldsymbol{y} - \boldsymbol{\Phi}^k\boldsymbol{x}\|_F^2 + \mu\|\boldsymbol{x} - \boldsymbol{z}^{k-1}\|_F^2 \tag{17}$$

$$\boldsymbol{z}^k = \underset{\boldsymbol{z}}{\text{argmin}} \frac{\mu}{2}\|\boldsymbol{z} - \boldsymbol{x}^k\|_F^2 + \tau R(\boldsymbol{z}) \tag{18}$$

where k indexes the iterative stages, and $\boldsymbol{\Phi}^k$ is the degradation matrix converted from the learned degradation $\widetilde{\mathcal{M}}^k$ in the kth stage. Then, these two subproblems can be solved alternatively. Equation (17) is a quadratic regularized least-squares problem that can be solved using a closed-form solution

$$\boldsymbol{x}^k = \left(\left(\boldsymbol{\Phi}^k\right)^T\boldsymbol{\Phi}^k + \mu\mathbf{I}\right)^{-1}\left(\left(\boldsymbol{\Phi}^k\right)^T\boldsymbol{y} + \mu\boldsymbol{z}^{k-1}\right) \tag{19}$$

where $\mathbf{I}$ is an identity matrix. $\boldsymbol{\Phi}^k$ is a large sparse matrix. As a result, it is computationally expensive to calculate the inverse of $(\boldsymbol{\Phi}^k)^T\boldsymbol{\Phi}^k + \mu\mathbf{I}$ directly. So, we follow the Woodbury matrix identity [46] to solve the inverse problem as

$$\left(\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \mu\mathbf{I}\right)^{-1} = \mu^{-1}\mathbf{I} - \mu^{-1}\boldsymbol{\Phi}^T\left(\mathbf{I} + \boldsymbol{\Phi}\mu^{-1}\boldsymbol{\Phi}^T\right)^{-1}\boldsymbol{\Phi}\mu^{-1}. \tag{20}$$

By substituting (20) into (19) and simplifying the equation, the closed-form solution of (17) is

$$\boldsymbol{x}^k = \boldsymbol{z}^{k-1} + \left(\boldsymbol{\Phi}^k\right)^T\left(\frac{\boldsymbol{y} - \boldsymbol{\Phi}^k\boldsymbol{z}^{k-1}}{\mu + \boldsymbol{\Phi}^k\left(\boldsymbol{\Phi}^k\right)^T}\right). \tag{21}$$

The solution of (18) can be regarded as a learnable proximal operator

$$\boldsymbol{z}^k = \text{Prox}\left(\boldsymbol{x}^k\right). \tag{22}$$

We design the DSST as the proximal operator.

1) Degradation-Guided Spectral–Spatial Transformer: Unlike many existing methods that only focus on spectral or spatial self-similarity, in this section, we propose DSST, which jointly captures the spectral and spatial dependencies under the guidance of the learned degradation.

As shown in Fig. 4(b), for the convenience of network implementation, we reshape the output of (21), $\boldsymbol{x}^k \in \mathbb{R}^{nC}$, to a tensor $\widetilde{\mathcal{X}}^k \in \mathbb{R}^{H\times(W+d(C-1))\times C}$, where $n = H\times(W+d(C-1))$, and then shift $\widetilde{\mathcal{X}}^k$ back to $\mathcal{X}^k \in \mathbb{R}^{H\times W\times C}$. To exploit the correlation between the reconstructed HSI and degradation, we concatenate $\mathcal{X}^k$ with a degradation-aware guidance map $\mathcal{O}_{\text{SL}} \in \mathbb{R}^{H\times W\times 1}$, which will be discussed below. A $\text{Conv}_{3\times 3}$

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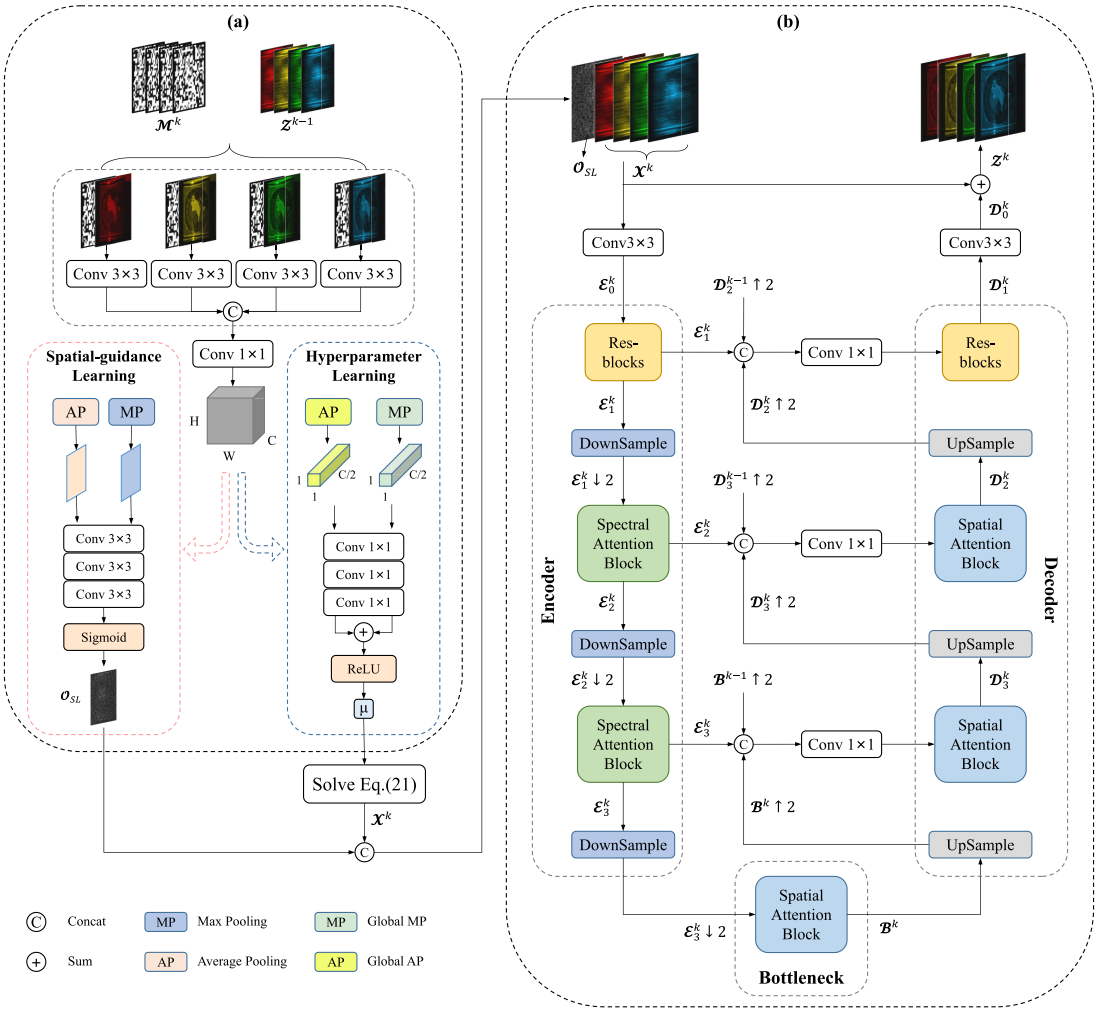


Fig. 4. Proximal operator of X-subproblem. (a) Structure of DL mechanism. (b) Structure of DSST.

Algorithm 1 Degradation-Aware Deep Prior Learning Network

Input: Measurement $\mathbf{Y}$
Output: Reconstructed HSI $\boldsymbol{\mathcal{X}}$ and degradation $\boldsymbol{\mathcal{M}}$
1: **Initialization:** $\boldsymbol{\mathcal{Z}}^0$, $\widetilde{\boldsymbol{\mathcal{M}}}_{coa}$ and $\widetilde{\boldsymbol{\mathcal{M}}}_{ref}^0$
2: **for** $k = 1, 2, \ldots, K$ **do:**
3: Solving $\widetilde{\mathcal{G}}_{ref}^k$ via (11)
4: Solving $\widetilde{\boldsymbol{\mathcal{M}}}_{ref}^k$ via (12)
5: $\widetilde{\boldsymbol{\mathcal{M}}}^k = \widetilde{\boldsymbol{\mathcal{M}}}_{coa} + \widetilde{\boldsymbol{\mathcal{M}}}_{ref}^k$
6: Solving $\boldsymbol{x}^k$ via (21)
7: Solving z^k via (22)
8: Convert z^k to $\widetilde{\boldsymbol{\mathcal{X}}}^k$
9: **end for**
10: $\boldsymbol{\mathcal{X}} \leftarrow \widetilde{\boldsymbol{\mathcal{X}}}^K$, $\boldsymbol{\mathcal{M}} \leftarrow \widetilde{\boldsymbol{\mathcal{M}}}^K$

(a convolution with a kernel size of 3×3) is used to map them into features $\boldsymbol{\mathcal{E}}_0^k$

$$\boldsymbol{\mathcal{E}}_0^k = \mathrm{Conv}_{3\times3}\big(\mathrm{Concat}\big(\boldsymbol{\mathcal{X}}^k, \boldsymbol{\mathcal{O}}_{\mathrm{SL}}\big)\big). \tag{23}$$

Afterward, we feed $\boldsymbol{\mathcal{E}}_0^k$ to the network. To extract the shallow feature and reduce the complexity, two res-blocks are used as the first level of the network. The second and third levels of the encoder are spectral self-attention blocks to capture the spectral dependencies. The bottleneck and the second and third levels of the decoder use spatial self-attention blocks to learn the spatial correlation. The upsampling and downsampling are implemented by a deconvolution with 2×2 kernel and a convolution with stride 4×4 kernel, respectively. To reduce the information loss, we adopt inner-stage and inter-stage information transfer strategies. Specifically, the features of encoder transfer to the same level of decoder in the same stage, and the features of bottleneck and the second and third levels of the decoder in the previous stage are upsampled and transferred to the next stage. As shown in Fig. 4(b), at the kth stage, the input of each decoder node consists of three components

$$\boldsymbol{\mathcal{DI}}_1^k = \mathrm{Conv}_{1\times1}\big(\mathrm{Concat}\big(\boldsymbol{\mathcal{D}}_2^k \uparrow 2,\ \boldsymbol{\mathcal{E}}_1^k,\ \boldsymbol{\mathcal{D}}_2^{k-1} \uparrow 2\big)\big) \tag{24}$$
$$\boldsymbol{\mathcal{DI}}_2^k = \mathrm{Conv}_{1\times1}\big(\mathrm{Concat}\big(\boldsymbol{\mathcal{D}}_3^k \uparrow 2,\ \boldsymbol{\mathcal{E}}_2^k,\ \boldsymbol{\mathcal{D}}_3^{k-1} \uparrow 2\big)\big) \tag{25}$$
$$\boldsymbol{\mathcal{DI}}_3^k = \mathrm{Conv}_{1\times1}\big(\mathrm{Concat}\big(\boldsymbol{\mathcal{B}}^k \uparrow 2,\ \boldsymbol{\mathcal{E}}_3^k,\ \boldsymbol{\mathcal{B}}^{k-1} \uparrow 2\big)\big) \tag{26}$$

where $\boldsymbol{\mathcal{DI}}_i^k$ is the input of the ith ($i = 1, 2, 3$) level of the decoder, $\uparrow 2$ means upsampling by 2 times, and $\boldsymbol{\mathcal{E}}_i^k$, $\boldsymbol{\mathcal{D}}_i^k$, and

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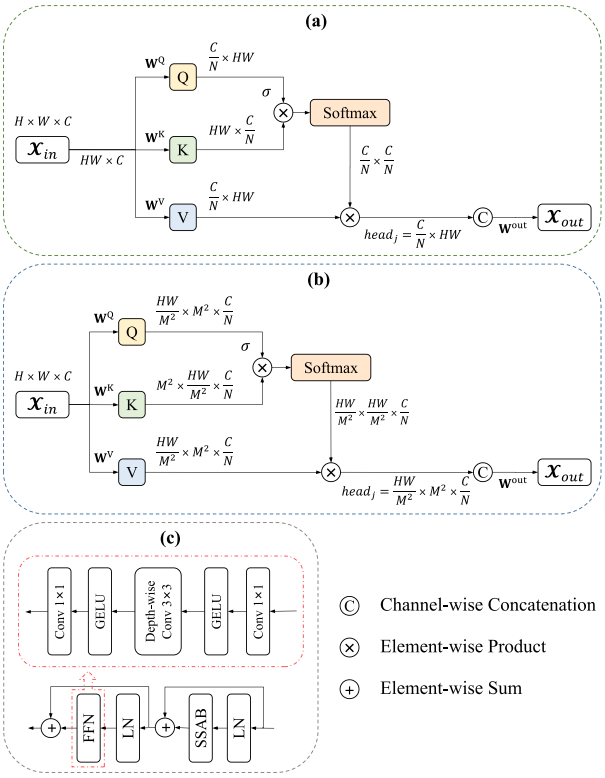


Fig. 5. Structure of the spectral/spatial attention block. (a) Calculation process of the spectral self-attention block. (b) Calculation process of the spatial self-attention block. (c) Structure shared by attention blocks. FFN is a feedforward network. LN is a layer normalization. SSAB refers to the spectral/spatial self-attention block shown in (a) and (b).

$\mathcal{B}^k$ are the outputs of the ith level of the encoder, decoder, and bottleneck, respectively.

Finally, we get features $\mathcal{D}_0^k$ after $\text{Conv}_{3\times3}$. With a long skip connection, the network outputs $\mathcal{Z}^k = \mathcal{X}^k + \mathcal{D}_0^k$.

As shown in Fig. 5(c), in DSST, both the spectral and spatial attention blocks share the same structure, which consists of two layer-normalization (LN) layers, a feedforward network (FFN), and a spectral/spatialwise self-attention block. The spectral self-attention block and spatial self-attention block are shown in Fig. 5(a) and (b), respectively.

a) Spectralwise Multihead Self-Attention Block: The spectralwise self-attention takes each spectral channel of input features as a token. Let the input of the spectral attention module be $\mathcal{X}_{\text{spe}}^{\text{in}} \in \mathbb{R}^{H\times W\times C}$. We reshape it into $\mathbf{X}_{\text{spe}} \in \mathbb{R}^{HW\times C}$, indicating that there are C tokens with a size of HW. Then, we transform $\mathbf{X}_{\text{spe}}$ into query $\mathbf{Q}_{\text{spe}} \in \mathbb{R}^{HW\times C}$, key $\mathbf{K}_{\text{spe}} \in \mathbb{R}^{HW\times C}$, and value $\mathbf{V}_{\text{spe}} \in \mathbb{R}^{HW\times C}$ by linear mapping

$$\mathbf{Q}_{\text{spe}} = \mathbf{X}_{\text{spe}}\mathbf{W}_{\text{spe}}^{Q} \quad \mathbf{K}_{\text{spe}} = \mathbf{X}_{\text{spe}}\mathbf{W}_{\text{spe}}^{K} \quad \mathbf{V}_{\text{spe}} = \mathbf{X}_{\text{spe}}\mathbf{W}_{\text{spe}}^{V} \tag{27}$$

where $\mathbf{W}_{\text{spe}}^{Q}$, $\mathbf{W}_{\text{spe}}^{K}$, $\mathbf{W}_{\text{spe}}^{V} \in \mathbb{R}^{C\times C}$ represent the linear mapping matrices. Then, we divide $\mathbf{Q}_{\text{spe}}$, $\mathbf{K}_{\text{spe}}$, $\mathbf{V}_{\text{spe}}$ into N heads along the channel dimension and get $\mathbf{Q}_{\text{spe}} = [\mathbf{Q}_{\text{spe}}^1, \mathbf{Q}_{\text{spe}}^2, \ldots, \mathbf{Q}_{\text{spe}}^N]$, $\mathbf{K}_{\text{spe}} = [\mathbf{K}_{\text{spe}}^1, \mathbf{K}_{\text{spe}}^2, \ldots, \mathbf{K}_{\text{spe}}^N]$, and $\mathbf{V}_{\text{spe}} = [\mathbf{V}_{\text{spe}}^1, \mathbf{V}_{\text{spe}}^2, \ldots, \mathbf{V}_{\text{spe}}^N]$. We calculate the spectralwise self-attention $\mathbf{A}_{\text{spe}}^j$ for each head $\mathbf{T}_{\text{spe}}^j$ as

$$\mathbf{A}_{\text{spe}}^j = \text{Softmax}\left(\sigma_{\text{spe}}^j \left(\mathbf{K}_{\text{spe}}^j\right)^T \mathbf{Q}_{\text{spe}}^j\right)$$
$$\mathbf{T}_{\text{spe}}^j = \mathbf{V}_{\text{spe}}^j \mathbf{A}_{\text{spe}}^j, \quad j \in \{1, 2, \ldots, N\} \tag{28}$$

where σ_{spe}^j is a learnable weight of self-attention. Subsequently, we concatenate the outputs of N heads along the channel dimension and use linear projection to map it to output $\mathcal{X}_{\text{spe}}^{\text{out}}$ as

$$\mathbf{X}_{\text{spe}} = \text{Concat}\left(\left[\mathbf{T}_{\text{spe}}^1, \mathbf{T}_{\text{spe}}^2, \ldots, \mathbf{T}_{\text{spe}}^N\right]\right)\mathbf{W}_{\text{spe}}^{\text{out}} \tag{29}$$

where $\mathbf{W}_{\text{spe}}^{\text{out}} \in \mathbb{R}^{C\times C}$ denotes the linear projection, and $\mathbf{X}_{\text{spe}} \in \mathbb{R}^{HW\times C}$ is the result of linear projection. Finally, we reshape $\mathbf{X}_{\text{spe}} \in \mathbb{R}^{HW\times C}$ into the final output $\mathcal{X}_{\text{spe}}^{\text{out}} \in \mathbb{R}^{H\times W\times C}$.

b) Spatialwise Multihead Self-Attention Block: We calculate the spatial attention in a window-based manner, which computes cross-window correlations to mine the nonlocal spatial dependencies. In addition, we let $\mathcal{X}_{\text{spa}}^{\text{in}} \in \mathbb{R}^{H\times W\times C}$ indicate the input and project it into query $\mathcal{Q}_{\text{spa}} \in \mathbb{R}^{H\times W\times C}$, key $\mathcal{K}_{\text{spa}} \in \mathbb{R}^{H\times W\times C}$, and value $\mathcal{V}_{\text{spa}} \in \mathbb{R}^{H\times W\times C}$. Then, we partition them into nonoverlapping windows with a size of $m \times m$ and reshape them to $\mathbb{R}^{(HW/m^2)\times m^2\times C}$. We divide the query, key, and value into N heads along the channel dimension. Then, we calculate the nonlocal spatial self-attention $\mathcal{A}_{\text{spa}}^j$ in each head $\mathcal{T}_{\text{spa}}^j$ as

$$\mathcal{A}_{\text{spa}}^j = \text{Softmax}\left(\sigma_{\text{spa}}^j \mathcal{Q}_{\text{spa}}^j \left(\mathcal{K}_{\text{spa}}^j\right)^T + \mathcal{P}_{\text{spa}}^j\right)$$
$$\mathcal{T}_{\text{spa}}^j = \mathcal{A}_{\text{spa}}^j \mathcal{V}_{\text{spa}}^j, \quad j = 1, 2, \ldots, N \tag{30}$$

where σ_{spa}^j is still a learnable parameter to reweight the attention, and $\mathcal{P}_{\text{spa}}^j \in \mathbb{R}^{(HW/m^2)\times(HW/m^2)\times(C/N)}$ are the learnable parameters to encode the position information. We concatenate all the heads and project them to get the output of the spatial attention block

$$\mathcal{X}_{\text{spa}} = \text{Concat}\left(\left[\mathcal{T}_{\text{spa}}^1, \mathcal{T}_{\text{spa}}^2, \ldots, \mathcal{T}_{\text{spa}}^N\right]\right)\mathbf{W}_{\text{spa}}^{\text{out}} \tag{31}$$

where $\mathbf{W}_{\text{spa}}^{\text{out}}$ is the linear projection matrix, and $\mathcal{X}_{\text{spa}} \in \mathbb{R}^{(HW/m^2)\times m^2\times C}$ records the result of (31). Finally, we reshape $\mathcal{X}_{\text{spa}}$ back to the final output $\mathcal{X}_{\text{spa}}^{\text{out}} \in \mathbb{R}^{H\times W\times C}$.

2) DL Mechanism: By emphasizing on the "black" pixels of coded aperture, the learned degradation could guide the DSST to better reconstruct HSI. In this section, we propose the DL mechanism, which could tell the network which areas on the image need more emphasis. In addition, it could also adaptively learn the hyperparameter μ in (21) instead of manually setting.

The inputs of DL module are $\mathcal{M}^k$, $\mathcal{Z}^{k-1} \in \mathbb{R}^{H\times W\times C}$ of intermediate stages. Each channel in the degradation tensor reveals the degradation pattern at the corresponding band, and the degradation patterns at different bands may not be identical. To match the degradation patterns with their corresponding bands and fully extract features, the degradation and HSI are integrated at the corresponding band in the DL. As shown in Fig. 4(a), each channel of the image $\mathcal{Z}^{k-1}$ and the degradation tensor $\mathcal{M}^k$ are grouped for feature extraction

$$\mathcal{D}_i = \text{Concat}\left(\mathcal{Z}_i^{k-1}, \mathcal{M}_i^k\right) \tag{32}$$
$$\mathcal{S} = [\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_C] \tag{33}$$

where $\mathcal{D}_i$ denotes the ith group, $i = 1, 2, \ldots, C$ indexes the groups, C is the channel number of $\mathcal{M}^k$ and $\mathcal{Z}^{k-1}$, and $\mathcal{S}$ denotes the set of all the groups. A $\text{Conv}_{3\times3}$ is used to

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extract features from each group and concatenate them along the channel dimension. Then, $\text{Conv}_{1\times1}$ is used to integrate the information

$$\mathcal{F} = \text{Conv}_{1\times1}(\text{Concat}(\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_C)) \tag{34}$$

where $\mathcal{F}_i = \text{Conv}_{3\times3}(\mathcal{D}_i)$. To reduce the computation and make the spatial-guidance learning (SL) and hyperparameter learning (HL) relative independently without disturbing each other, we divide $\mathcal{F}$ into two parts, $\mathcal{F}_{\text{SL}}$ and $\mathcal{F}_{\text{HL}}$, along the channel, for SL and HL, respectively.

a) Spatial-Guidance Learning: SL learns a degradation guidance map telling the network which areas on the images need emphasis. We apply the average and maximum pooling along the channel. Features are extracted from the pooled features through three $\text{Conv}_{3\times3}$. Activated by the Sigmoid function, a degradation guidance map can be obtained

$$\mathcal{R}_{\text{ave}} = \text{avg_pool}(\mathcal{F}_{\text{SL}}) \tag{35}$$

$$\mathcal{R}_{\text{max}} = \text{max_pool}(\mathcal{F}_{\text{SL}}) \tag{36}$$

$$\mathcal{O}_{\text{SL}} = \text{Sigmoid}(\text{Conv}_{3\times3}(\text{Concat}(\mathcal{R}_{\text{ave}}, \mathcal{R}_{\text{max}}))) \tag{37}$$

where avg_pool$(\cdot)$ and max_pool$(\cdot)$ denote the average and maximum pooling along the channel dimension, respectively. The final output $\mathcal{O}_{\text{SL}} \in \mathbb{R}^{H\times W\times1}$ is fed to the DSST to provide degradation guidance.

b) Hyperparameter Learning: HL learns the penalty parameter μ in (21). It performs global average and maximum pooling on the input, and then reduces the number of channels to one through shared fully connected (FC) layers. We use the ReLU function to activate it

$$\mathcal{R}_{\text{gav}} = \text{glb_avg}(\mathcal{F}_{\text{HL}}) \tag{38}$$

$$\mathcal{R}_{\text{gma}} = \text{glb_max}(\mathcal{F}_{\text{HL}}) \tag{39}$$

$$\mathcal{O}_{\text{HL}} = \text{ReLU}\big(\text{FC}\big(\mathcal{R}_{\text{gav}}\big) + \text{FC}\big(\mathcal{R}_{\text{gma}}\big)\big) \tag{40}$$

where glb_avg$(\cdot)$ and glb_max$(\cdot)$ denote the global average and maximum pooling, respectively. The final output $\mathcal{O}_{\text{HL}} \in \mathbb{R}^{1\times1\times1}$ acts as the parameter μ in (21).

3) Training Loss: To learn the degradation and latent HSI, we propose a joint loss function measuring the HSI reconstruction errors, degradation fidelity, and degradation consistency.

E. Reconstruction Loss L_{rec}

The reconstruction loss L_{rec} measures the reconstructed errors of the inferred HSI

$$L_{\text{rec}} = \frac{1}{B}\sum_{i=1}^{B}\left\|\mathcal{X}_{\text{gt}}^{i} - \mathcal{X}_{\text{rec}}^{i}\right\|_F^2 \tag{41}$$

where B indexes the training samples, $\mathcal{X}_{\text{gt}}^{i}$ represents the ground-truth (GT) image of the ith sample, and $\mathcal{X}_{\text{rec}}^{i}$ denotes the reconstructed result of the ith sample. The reconstruction loss ensures that the reconstructed HSI is close to the real HSI.

1) Degradation-Fidelity Loss L_{df}: The degradation-fidelity loss L_{df} is

$$L_{\text{df}} = \frac{1}{B}\sum_{i=1}^{B}\left\|\sum_{j=1}^{C}\widetilde{\mathcal{M}}_{\text{coa}}^{i}(:,:,j)\odot\widetilde{\mathcal{X}}_{\text{gt}}^{i}(:,:,j) - \mathbf{Y}_i\right\|_F^2 \tag{42}$$

where C indexes the channels, $\widetilde{\mathcal{M}}_{\text{coa}}^{i}$ denotes the estimated coarse degradation obtained by the CDE in Section III-C, and $\mathbf{Y}_i$ is the measurement of the ith samples. L_{df} ensures that the learned coarse degradation generally approximates the physical coded aperture of CASSI.

2) Degradation-Consistency oss L_{dc}: The degradation-consistency loss L_{dc} modulates the reconstructed image $\widetilde{\mathcal{X}}_{\text{rec}}$ using the learned degradation $\widetilde{\mathcal{M}}$ to generate the measurements and requires them to be consistent with the real measurements

$$L_{\text{dc}} = \frac{1}{B}\sum_{i=1}^{B}\left\|\sum_{j=1}^{C}\Big(\widetilde{\mathcal{M}}_{\text{coa}}^{i}(:,:,j) + \widetilde{\mathcal{M}}_{\text{ref}}^{i}(:,:,j)\Big)\odot\widetilde{\mathcal{X}}_{\text{rec}}^{i}(:,:,j) - \mathbf{Y}_i\right\|_F^2. \tag{43}$$

Compared with L_{df}, L_{dc} jointly constrains $\widetilde{\mathcal{M}}_{\text{coa}}$, $\widetilde{\mathcal{M}}_{\text{ref}}$, and the reconstructed HSI, which ensures that the final learned degradation and reconstructed image are consistent with the real degradation and measurement.

The joint loss function is

$$L_{\text{joint}} = L_{\text{rec}} + \delta_1 L_{\text{df}} + \delta_2 L_{\text{dc}} \tag{44}$$

where δ_1 and δ_2 are the weights to balance different losses. We manually set $\delta_1 = 0.001$ and $\delta_2 = 0.0005$.

IV. Experiments

A. Experimental Settings

1) Simulation Dataset: We conduct the simulation experiments on three public HSI datasets, the CAVE dataset [47], the KAIST dataset [48], and the ICVL dataset [49]. The CAVE dataset consists of 32 scenes with spatial size 512×512. The KAIST dataset has 30 scenes with spatial size 2704×3376. There are nearly 200 HSIs with a resolution of 1392×1300 in the ICVL dataset. In these datasets, each HSI has 31 spectral bands. Through spectral interpolation, for the CAVE and KAIST datasets, we adopt 28 wavelengths ranging from 450 to 650 nm; and for the ICVL dataset, we adopt 24 wavelengths ranging from 400 to 700 nm. We conduct two simulation experiments.

1) Sim. 1: Trained on the CAVE dataset and tested on the KAIST dataset.
2) Sim. 2: Trained and tested on the ICVL dataset.

In Sim. 1, we take the CAVE dataset as the training dataset and select ten scenes from the KAIST dataset for testing as shown in Fig. 6. We crop all the HSIs into patches with size $256\times256\times28$. In Sim. 2, we select 80 images from the ICVL dataset as the training set and another ten images for testing as shown in Fig. 7. In addition, we crop these HSIs into patches with a size of $256\times256\times24$. We use a fixed mask with size 256×256 to simulate the 2-D measurements by setting the shifting step size d to 2.

2) Real Dataset: Other than the simulation experiments, we also test different methods on real CASSI data. Five real measurements captured by the CASSI are used as testing data [26], as shown in Fig. 8. These five real measurements are

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TABLE I

QUANTITATIVE RESULTS ON TEN KAIST TESTING SCENES (S1–S10) OF SIM. 1. PARAMS, FLOPS, PSNR (UPPER ENTRY IN EACH CELL), AND SSIM (LOWER ENTRY IN EACH CELL) ARE LISTED. THE BEST RESULTS ARE IN RED AND THE SECOND-BEST RESULTS ARE IN BLUE

Methods		Params	FLOPs	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Avg
GAP-TV [25]	ICIP 2016	–	–	27.15 0.767	26.00 0.691	26.91 0.799	35.40 0.910	24.65 0.737	23.72 0.693	24.69 0.717	23.24 0.663	25.49 0.746	24.70 0.621	26.20 0.734
DeSCI [23]	TPAMI 2018	–	–	27.85 0.794	26.30 0.699	28.43 0.858	38.09 0.949	25.30 0.773	24.07 0.750	25.43 0.772	23.42 0.735	26.52 0.797	24.74 0.663	27.05 0.779
ADMM-Net [34]	ICCV 2019	4.27M	84.89G	33.68 0.915	32.73 0.883	34.00 0.916	40.45 0.962	31.27 0.911	32.09 0.916	31.83 0.885	30.31 0.894	32.27 0.899	30.07 0.885	32.87 0.907
λ-Net [27]	ICCV 2019	32.73M	32.17G	32.55 0.895	31.66 0.861	34.25 0.934	41.01 0.967	30.07 0.896	30.74 0.900	30.73 0.885	29.56 0.889	32.48 0.896	28.80 0.845	32.19 0.897
TSA-Net [26]	ECCV 2020	44.25M	135.17G	33.15 0.916	32.12 0.896	33.34 0.933	40.49 0.972	30.42 0.915	32.03 0.932	31.20 0.897	30.33 0.920	31.71 0.921	29.85 0.892	32.47 0.919
DGSMP [31]	CVPR 2021	3.76M	647.79G	33.85 0.930	33.44 0.926	33.94 0.941	39.22 0.969	30.71 0.932	33.29 0.941	32.16 0.913	31.57 0.937	31.74 0.926	31.71 0.939	33.16 0.935
HerosNet [33]	CVPR 2022	11.82M	447.55G	35.12 0.936	34.27 0.924	34.75 0.923	40.77 0.964	32.34 0.933	34.51 0.947	33.69 0.923	32.61 0.937	34.17 0.927	32.28 0.931	34.45 0.934
MST [28]	CVPR 2022	2.02M	27.48G	35.23 0.944	35.71 0.946	36.66 0.956	42.94 0.979	32.41 0.948	34.67 0.957	33.43 0.927	32.99 0.956	35.12 0.949	32.50 0.947	35.17 0.951
CST [30]	ECCV 2022	3.00M	27.19G	35.11 0.943	35.46 0.943	36.36 0.956	43.08 0.979	32.72 0.949	35.08 0.960	33.58 0.929	33.40 0.958	35.10 0.949	32.70 0.948	35.26 0.951
Ours	–	7.65M	84.71G	35.83 0.949	36.97 0.956	37.72 0.961	43.93 0.983	33.44 0.957	35.59 0.965	34.54 0.938	33.44 0.958	36.43 0.954	33.04 0.950	36.09 0.957



Fig. 6. Ten testing scenes from the KAIST dataset in Sim. 1 experiment.

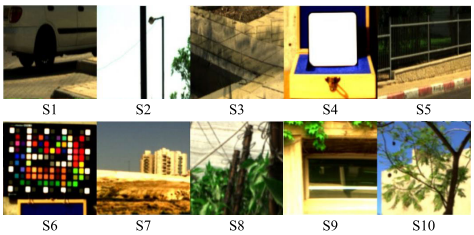


Fig. 7. Ten testing scenes from the ICVL dataset in Sim. 2 experiment.



Fig. 8. RGB images of five real measurements captured by CASSI [26].

modulated by a real mask of size 660×660, and the shifting step is 2. Therefore, the size of each measurement is 660×714. The KAIST dataset is used to train the HSI reconstruction networks. To match the size of the CASSI measurements, we randomly crop patches with spatial size 660×660 from the KAIST dataset as training samples and use the real mask to modulate them. Simulating the reality, 11-bit shot noise is added to the training data.

3) Implementation Details: Our model is developed on PyTorch, and all the networks are trained and tested on the NVIDIA RTX 3090 GPU. We use the Adam [50] optimizer ($\beta_1 = 0.9$ and $\beta_2 = 0.999$) to train our model for 200 epochs and initialize the learning rate to 0.0004 decayed by 0.5 every 50 epochs. The batch size is set to 5. We unfold the network with three iterative stages, and the parameters of each stage in our unfolding framework are shared except for the DSST module.

B. Simulation Results

Several commonly used metrics, such as peak signal-to-noise ratio (PSNR) [51], SSIM [51], spectral angle mapper (SAM) [52], RMSE, erreur relative globale sdimensionnelle de synthèse (ERGAS) [53], and spatial correlation coefficient (SCC) [54], are adopted to measure the performance. And we calculate the number of parameters (Params) and floating point operations (FLOPs) to evaluate the efficiency of different methods. We compare our method with the SOTA algorithms, including model-based methods (GAP-TV [25], DeSCI [23]) and learning-based algorithms (ADMM-Net [34], λ-Net [27], TSA-Net [26], MST [26], CST [30], DGSMP [31], HerosNet [33]).

1) Numerical Results: In Tables I and II, we list the Params, FLOPs, PSNR, and SSIM results of each method. The average PSNR and SSIM results of each method are shown in the rightmost column of the tables. In Tables III and IV, the average of another four metrics, SAM, RMSE, ERGAS, and SCC, are listed. The best results are in red, and the second-best results are in blue.

From Tables I to IV, the metrics by our method in Sim. 1 and Sim. 2 experiments are competitive. Specifically, for the

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TABLE II

QUANTITATIVE RESULTS ON TEN ICVL TESTING SCENES (S1–S10) OF SIM. 2. PARAMS, FLOPS, PSNR (UPPER ENTRY IN EACH CELL) AND SSIM (LOWER ENTRY IN EACH CELL) ARE LISTED. THE BEST RESULTS ARE IN RED AND THE SECOND-BEST RESULTS ARE IN BLUE

Methods		Params	FLOPs	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Avg
GAP-TV [25]	ICIP 2016	–	–	29.62 0.833	20.08 0.726	25.13 0.770	24.54 0.803	28.13 0.758	22.20 0.642	28.46 0.843	23.62 0.670	31.28 0.909	22.47 0.623	25.55 0.758
DeSCI [23]	TPAMI 2018	–	–	31.67 0.892	20.75 0.751	25.71 0.774	26.24 0.854	29.36 0.795	23.02 0.719	29.81 0.870	23.97 0.670	33.16 0.946	22.83 0.625	26.65 0.790
ADMM-Net [34]	ICCV 2019	4.27M	84.89G	33.14 0.874	27.15 0.829	31.02 0.867	28.61 0.844	31.11 0.796	25.46 0.751	32.50 0.906	28.27 0.795	32.03 0.833	24.29 0.708	29.36 0.820
λ-Net [27]	ICCV 2019	32.73M	32.17G	33.63 0.854	29.72 0.913	33.42 0.902	30.61 0.891	31.91 0.805	28.90 0.830	33.34 0.919	30.24 0.846	32.06 0.812	26.20 0.773	31.01 0.854
TSA-Net [26]	ECCV 2020	44.25M	135.17G	36.09 0.915	29.62 0.920	33.50 0.915	31.80 0.911	34.24 0.865	29.55 0.863	34.81 0.931	30.75 0.858	34.68 0.893	26.41 0.777	32.15 0.885
DGSMP [31]	CVPR 2021	3.76M	647.79G	35.00 0.895	30.02 0.929	34.04 0.917	29.57 0.890	33.81 0.853	28.87 0.845	34.48 0.932	31.28 0.870	33.44 0.859	28.04 0.829	31.86 0.882
HerosNet [33]	CVPR 2022	11.82M	447.55G	37.03 0.916	31.52 0.921	35.53 0.938	30.45 0.876	35.09 0.865	28.87 0.842	35.95 0.938	32.84 0.889	33.56 0.869	27.95 0.819	32.88 0.887
MST [28]	CVPR 2022	2.02M	27.48G	35.66 0.904	31.69 0.933	35.07 0.926	33.01 0.908	33.77 0.845	30.66 0.864	35.17 0.940	32.04 0.880	33.25 0.850	28.03 0.822	32.83 0.887
CST [30]	ECCV 2022	3.00M	27.19G	34.76 0.881	30.76 0.938	34.62 0.921	32.06 0.908	34.02 0.856	30.48 0.864	34.87 0.936	31.64 0.874	33.21 0.851	27.69 0.821	32.41 0.885
Ours	–	7.65M	84.71G	37.79 0.926	32.32 0.950	36.50 0.938	34.09 0.931	36.03 0.894	31.96 0.895	37.47 0.954	33.02 0.896	35.64 0.901	28.40 0.837	34.32 0.912

TABLE III

SAM, RMSE, ERGAS, AND SCC RESULTS OF SIM. 1

Methods	SAM	RMSE	ERGAS	SCC
GAP-TV [25]	0.292	0.052	67.60	0.877
DeSCI [23]	0.245	0.049	62.31	0.893
ADMM-Net [34]	0.218	0.024	31.67	0.973
λ-Net [27]	0.207	0.026	34.55	0.967
TSA-Net [26]	0.130	0.025	33.36	0.971
DGSMP [31]	0.149	0.023	32.68	0.976
HerosNet [33]	0.179	0.020	27.37	0.981
MST [28]	0.127	0.019	24.42	0.984
CST [30]	0.117	0.018	24.74	0.984
Ours	0.111	0.017	22.63	0.987

TABLE IV

SAM, RMSE, ERGAS, AND SCC RESULTS OF SIM. 2

Methods	SAM	RMSE	ERGAS	SCC
GAP-TV [25]	0.121	0.058	56.41	0.883
DeSCI [23]	0.105	0.053	45.06	0.894
ADMM-Net [34]	0.163	0.037	48.64	0.962
λ-Net [27]	0.169	0.030	46.39	0.976
TSA-Net [26]	0.104	0.027	29.57	0.981
DGSMP [31]	0.147	0.027	48.11	0.983
HerosNet [33]	0.139	0.025	43.40	0.978
MST [28]	0.148	0.024	43.94	0.971
CST [30]	0.141	0.026	44.95	0.985
Ours	0.102	0.021	27.33	0.985

iterative algorithms, GAP-TV and DeSCI, they take a long time to manually tune the parameters to achieve better results, but their results are still unsatisfactory. The main reason may be that the hand-crafted priors are not sufficient to fully capture the priors of HSI. DUNs learn deep priors by neural networks. ADMM-Net, DGSMP, and HerosNet get better results than the model-based methods. ADMM-Net simply uses a U-Net to learn the deep prior, without considering the inherent characteristics of HSI, leading to limited performance. λ-Net and CST propose to learn spatial dependencies, whereas MST captures spectral dependencies under the guidance of mask. Only spectral or spatial dependencies may not be sufficient. TSA-Net learns both spatial and spectral correlations, but the results may be oversmoothed. By learning a reliable degradation and capturing the dependencies in the spectral and spatial domains under the guidance of degradation, our method achieves the state-of-the-art results with an acceptable number of parameters and computational effort.

2) Visualization of Testing Scenes: We visualize the reconstructed results in Figs. 9 and 10. Four channels are randomly selected from each HSI for comparison. We can find that the reconstructed images obtained by deep-learning-based methods are generally better than model-based methods. The images obtained by our method are closer to the GT, and it could reconstruct richer textures and sharper edges. TSA-Net takes a large number of parameters to capture the spatial–spectral information of the images, and the reconstructed images are oversmoothed. CST achieves the second-best results, but it inevitably loses spatial information due to the coarse patch selecting mechanism, which only learns the dependencies of selected region in the spatial domain.

Furthermore, to verify the spectral consistency, we select two scenes from the testing sets and arbitrarily select two small areas from them, as framed by the colored boxes in Fig. 11. We plot the spectral density curves and list the correlation coefficient, which is obtained by performing global average pooling on the framed area of the reconstructed image and GT and calculating their inner product. The larger the value,

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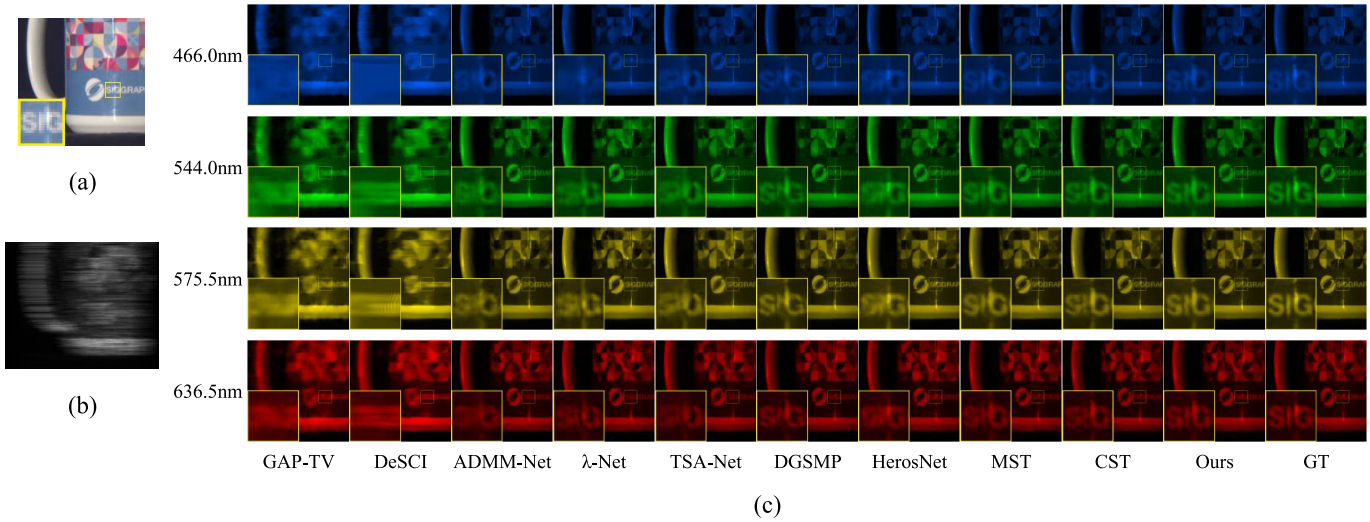


Fig. 9. Visual comparison of HSI reconstruction results in Sim. 1 experiment. The testing image is from the KAIST dataset (scene 5) and four spectral bands are presented. (a) RGB image of the KAIST testing image. (b) Measurement of the KAIST testing image. (c) Visualization of the reconstructed HSI and four spectral bands are presented.

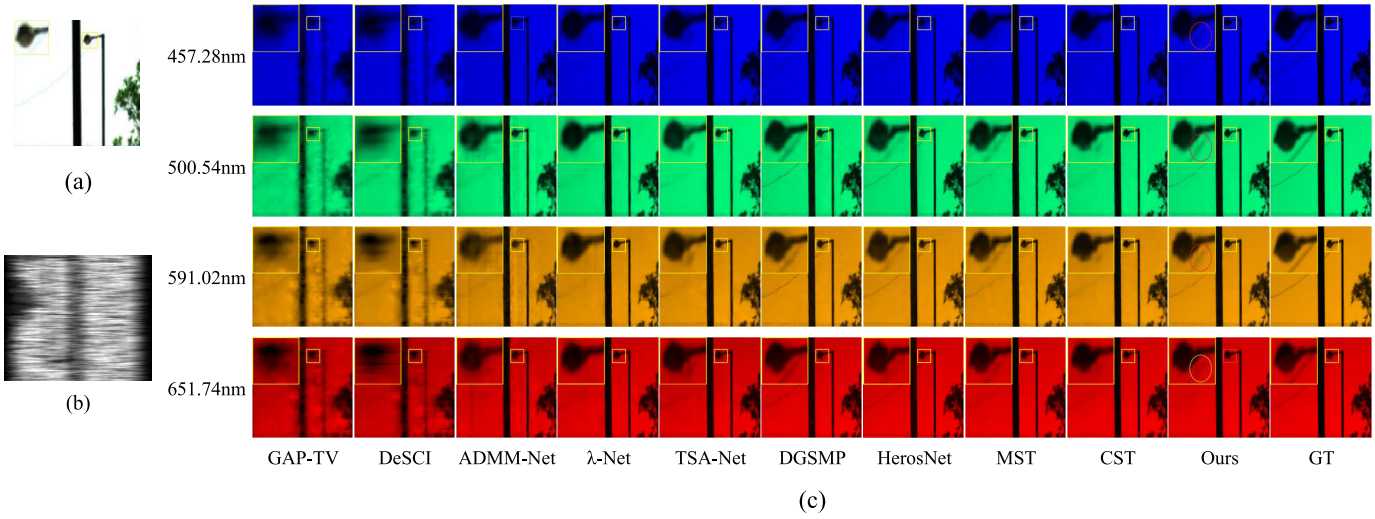


Fig. 10. Visual comparison of HSI reconstruction results in Sim. 2 experiment. The testing image is from the ICVL dataset (scene 2) and four spectral bands are presented. (a) RGB image of the ICVL testing image. (b) Measurement of the ICVL testing image. (c) Visualization of the reconstructed HSI and four spectral bands are presented.

the stronger the correlation. These results demonstrate that our method leads to better spectral consistency.

C. Real Data Results

We train the HSI reconstruction networks on the KAIST dataset and apply them on five real measurements captured by real CASSI [26]. The reconstructed HSIs by different networks are represented in Fig. 12. We can easily observe that our method can restore more details with clearer textures and sharper edges. Compared with other methods, the proposed degradation learning mechanism predicts the degradation that is close to the real one, which mitigates the distortion problem in real imaging process. And the degradation-awareness guides the learning of spectral and spatial dependencies, which assists the network to emphasize on the "black" pixels of the coded aperture and improves HSI reconstruction. Benefiting from the learning of degradation, our method has a significant improvement over others in reconstructing HSI from real measurements.

D. Ablation Study

In this section, we conduct the ablation experiments on the CAVE and KAIST datasets. As Sim. 1, we train the models on the CAVE dataset and test on the KAIST dataset.

1) CDR and DSST Ablation: This ablation aims to study the CDR mechanism (**M**-subproblem) and DSST (**X**-subproblem). We break down our network to baseline-1 by maintaining the unfolding framework, while replacing the transformer with a three-level U-Net and removing CDR. As listed in Table V, baseline-1 gets 29.58 dB in PSNR and 0.731 in SSIM. When we apply the CDR to cases 1 and 3, the PSNR improves by 3.9 and 0.87 dB, respectively. It shows that our CDR

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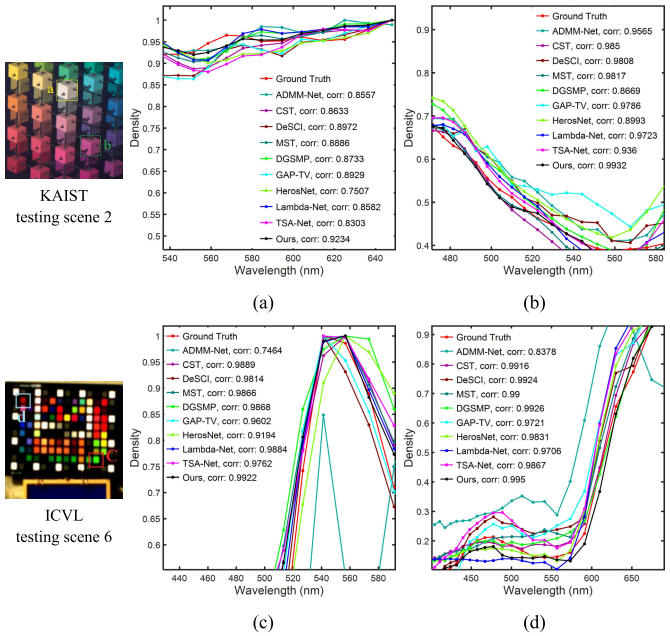


Fig. 11. Spectral density curves of two testing images from the KAIST and ICVL datasets, respectively. (a)–(d) Spectral density curves of the colored box in the RGB images, respectively.

TABLE V

ABLATION OF CONTENT-AWARE DEGRADATION REFINEMENT AND DEGRADATION-GUIDED SPECTRAL–SPATIAL TRANSFORMER

Cases	Baseline-1	CDR	DSST	PSNR	SSIM
case 1	✓			29.58	0.731
case 2	✓	✓		33.48	0.901
case 3	✓		✓	35.22	0.950
case 4	✓	✓	✓	36.09	0.957

TABLE VI

ABLATION OF LOSS FUNCTION

Cases	L_{rec}	L_{df}	L_{dc}	PSNR	SSIM
case 1	✓			35.08	0.949
case 2	✓	✓		35.24	0.951
case 3	✓		✓	35.04	0.948
case 4	✓	✓	✓	36.09	0.957

mechanism could narrow the gap between the ideal and real degradations, which improves the reconstructed HSI. When we apply the DSST in cases 1 and 2, the PSNR improves by 5.64 and 2.61 dB, respectively. It demonstrates the ability of DSST in helping the HSI reconstruction.

2) Loss Function Ablation: We conduct ablation of loss function on D^{2}PL-Net in Table VI. The reconstruction loss L_{rec} is the main part of the loss function; it measures the errors between the reconstructed image and the GT. We analyze the impact of L_{df} and L_{dc}. The degradation-fidelity loss L_{df} ensures that the estimated coarse degradation matrix would be consistent with the coded aperture of CASSI. Without it, the estimated matrix would be random and unordered. The comparison between cases 1, 2 and cases 3, 4 verifies the effectiveness of L_{df}. L_{dc} requires the HSI modulated by learned degradation to be consistent with the real measurement captured by CASSI. Without L_{dc}, the value of degradation matrix would be uncontrollable and large. In general, the ablation study in Table VI demonstrates the necessity of the joint losses.

TABLE VII

ABLATION OF SELF-ATTENTION MECHANISM. SPE AND SPA DENOTE SPECTRAL ATTENTION AND SPATIAL ATTENTION. ENC-SPE/SPA DENOTES THE ENCODER OF THE TRANSFORMER LEARNS SPECTRAL/SPATIAL ATTENTION, WHILE DEC-SPE/SPA DENOTES THE DECODER OF THE TRANSFORMER LEARNS SPECTRAL/SPATIAL ATTENTION

Cases		PSNR	SSIM
case 1	Only SPE	35.06	0.949
case 2	Only SPA	34.41	0.943
case 3	Enc-SPA and Dec-SPE	34.78	0.947
case 4	Enc-SPE and Dec-SPA	35.22	0.950

TABLE VIII

ABLATION OF DEGRADATION-GUIDANCE LEARNING MECHANISM. HL REPRESENT THE HYPERPARAMETER LEARNING BRANCH. AND SL IS THE SPATIAL-GUIDANCE LEARNING BRANCH

Cases	Baseline-1	HL	SL	PSNR	SSIM
case 1	✓			29.58	0.731
case 2	✓	✓		29.64	0.732
case 3	✓		✓	31.85	0.864
case 4	✓	✓	✓	32.15	0.867

TABLE IX

ABLATION OF UNFOLDING STAGES (K)

Cases	Stages	Params	FLOPs	PSNR	SSIM
case 1	$K=2$	5.83M	56.89G	35.01	0.948
case 2	$K=3$	7.65M	84.71G	36.09	0.957
case 3	$K=4$	9.48M	112.96G	35.75	0.956
case 4	$K=5$	11.31M	141.21G	35.96	0.957

3) Self-Attention Mechanism Ablation: To verify the combination of spectral and spatial self-attention mechanisms, we construct four different transformers. As shown in Table VII, the most effective combination of these two self-attention mechanisms is learning spectral attention in the encoder and spatial attention in the decoder (case 4), which enables our transformer to achieve the highest results, 35.22 dB in PSNR and 0.950 in SSIM.

4) DL Mechanism Ablation: This ablation aims to study the DL module, which contains HL and SL branches. As shown in Table VIII, the SL branch helps the model gain a 2.27-dB improvement in PSNR. Integrated with HL branch, it could further improve HSI reconstruction.

5) Unfolding Stages Ablation: This ablation studies the performance of different unfolding stages K of our method. As listed in Table IX, when $K=3$, our method achieves the best performance. As K continues to increase, the performance plateaus, so we set $K=3$.

E. Visualization Analysis

1) Visualization of Spatial Maps of the Transformer: We visualize the spatial maps of the transformer in Fig. 13 to study why learning spectral attention in the encoder and spatial attention in the decoder is efficient. From these visualizations, we can see that the decoders usually contain richer and clearer

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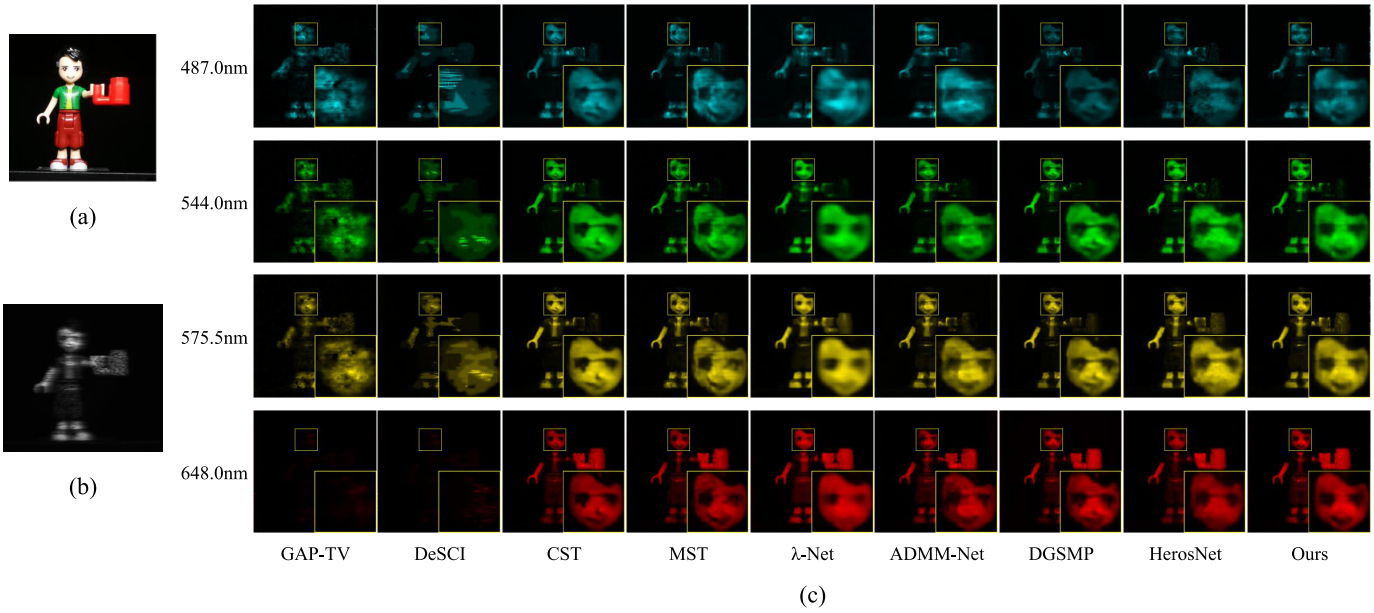


Fig. 12. Visual comparison of real HSI reconstruction results. The testing image is from real measurement (scene 3) and four spectral channels are presented. (a) RGB image of the testing image. (b) Measurement of the testing image. (c) Visualization of the reconstructed HSI and four spectral bands are presented.

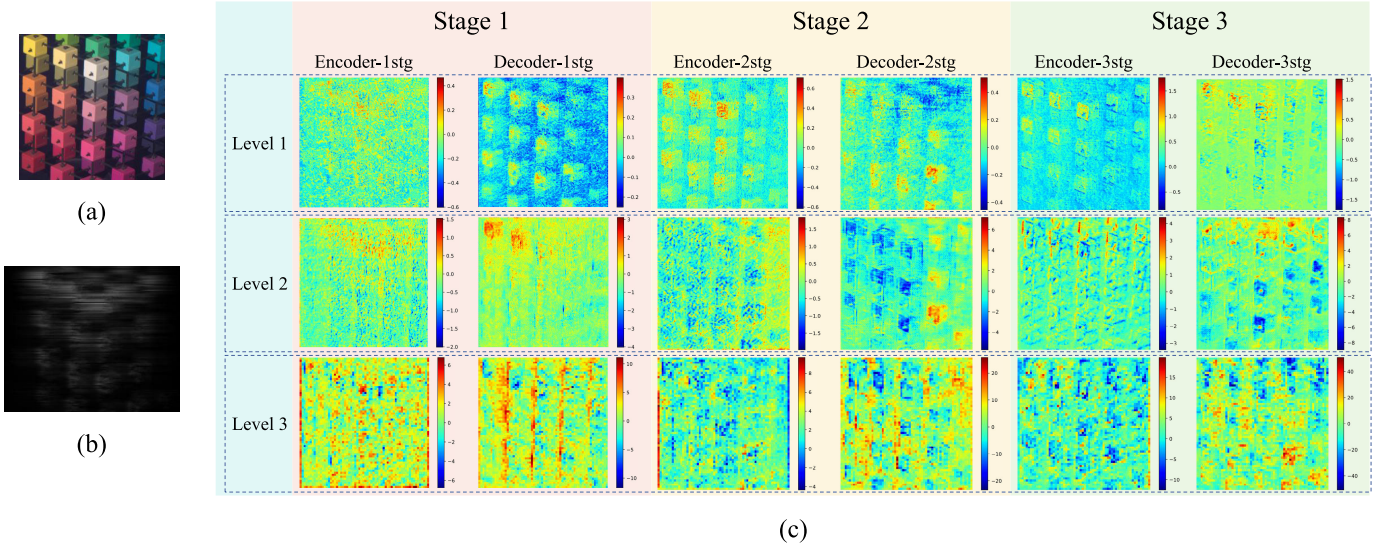


Fig. 13. Visual comparison of spatial maps of the transformer in different unfolding stages. (a) RGB image of the KAIST testing scene 2. (b) Measurement of the testing image. (c) Visualization of spatial information in the three unfolding stages.

spatial information than the encoders in every stage. The reason might be that the unique sampling process of the CASSI causes aliasing between features in the encoder part, which is not conducive to capturing spatial dependencies compared with the decoder part.

2) Visualization of Spatial Guidance Map: In Fig. 14, we visualize the spatial guidance map $\mathcal{O}_{\text{SL}}$ learned by the DL mechanism. From these visualizations, we can see that the spatial guidance maps contain the degradation information, which may guide the network to focus on the "black" areas modulated by the mask.

3) Visualization of DML: We visualize the learned degradation matrix for the KAIST testing and real testing scene 3 in Fig. 15. From these visualizations, we can find that the CDE mechanism can estimate a reliable coarse degradation $\mathbf{M}_{\text{coa}}$ that is close to GT degradation. We use the root mean square error (RMSE) to measure the similarity between the learned degradation $\mathbf{M}$ and the GT degradation. The RMSE of the KAIST testing scene 3 is 0.435, while the RMSE of the real testing scene 3 is 0.704. And the CDR mechanism can learn the refinement component $\mathbf{M}_{\text{ref}}$ for each pixel of degradation

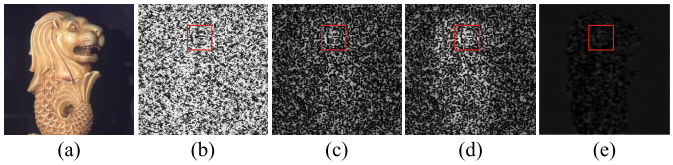


Fig. 14. Visualization of the learned spatial guidance map $\mathcal{O}_{\text{SL}}$. (a) RGB image of the KAIST testing scene 10. (b) Corresponding degradation. (c)–(e) Spatial guidance maps learned in the first, second, and third stages of the unfolding network, respectively.

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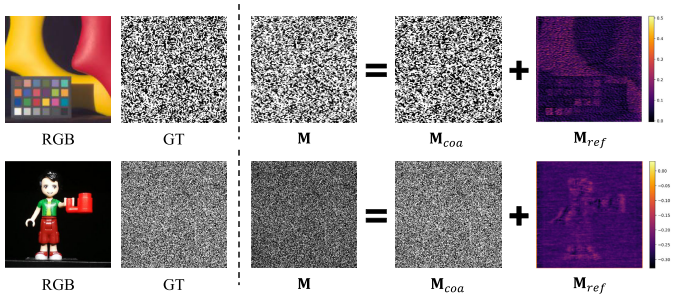


Fig. 15. Visualizations of the refined degradation $\mathbf{M}$ obtained by adding $\mathbf{M}_{\text{coa}}$ and $\mathbf{M}_{\text{ref}}$. The upper row is the visualized degradation of the KAIST testing scene 3. The lower is the visualization of real testing scene 3. The RMSE between GT and $\mathbf{M}$ of the KAIST testing scene 3 is 0.435, while the RMSE of the real testing scene 3 is 0.704.

matrix. Even the value at each pixel of learned refinement is relatively small, and it could refine the coarse degradation to make it more approximate to the real degradation of CASSI.

V. Conclusion

In this study, we propose D^{2}PL-Net, which integrates the degradation learning and HSI reconstruction into a unified framework. Considering the phase aberration and distortion problems in the real imaging process, there is a certain gap between the real and ideal degradations. So, we propose to learn a proper degradation by CDE and CDR mechanisms. Then, to reconstruct the high-quality HSI, we propose DSST to learn the deep prior for HSI reconstruction. The DSST is proposed to simultaneously capture the long-range dependencies in the spectral and spatial domains of HSI under the guidance of degradation. From the experimental analysis, we can conclude that our D^{2}PL-Net is effective. However, since our framework simultaneously learns the degradation and reconstructs HSI, compared with other methods that only involve HSI reconstruction without degradation learning, the computation would be a little larger. How to reduce the computation would be the future work.

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